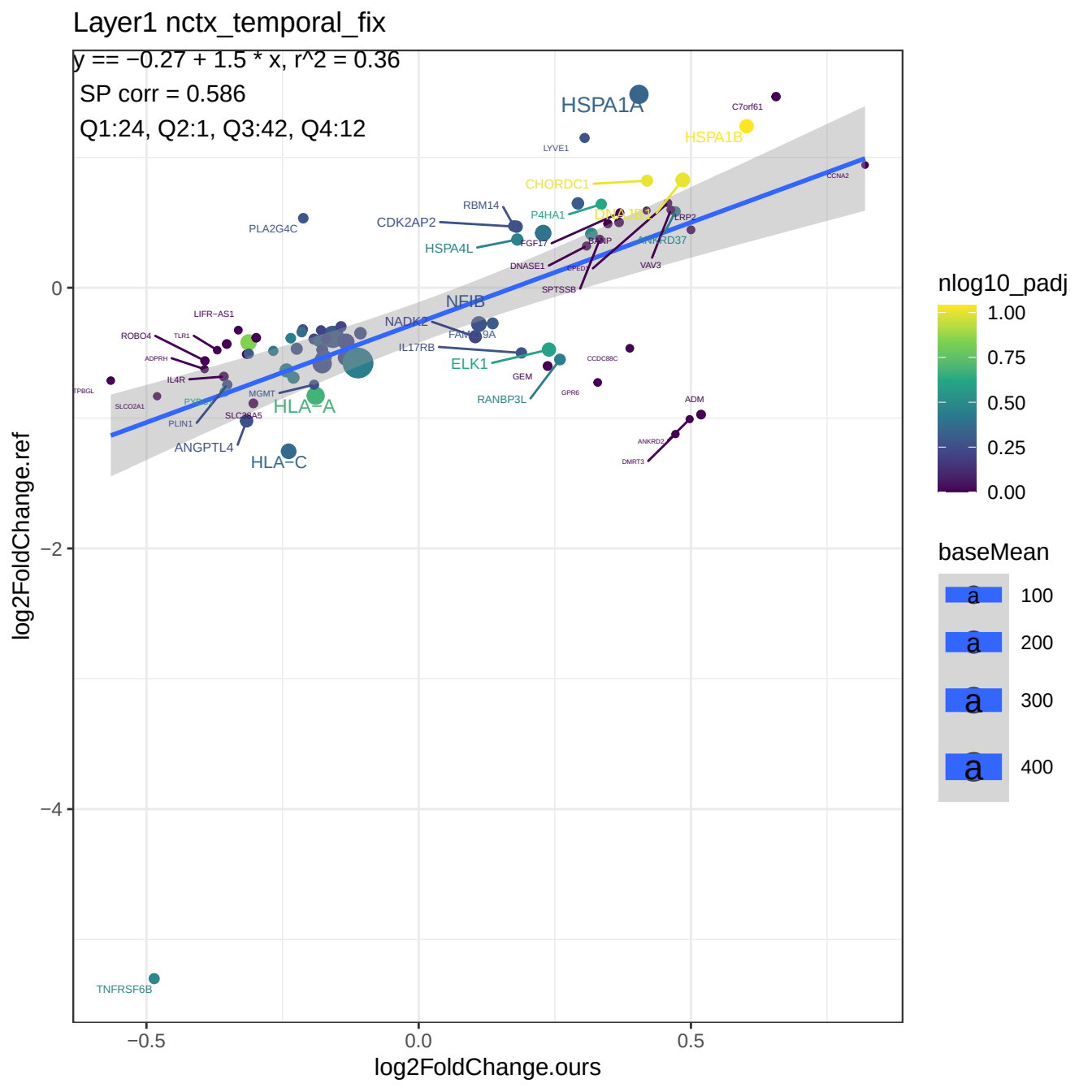
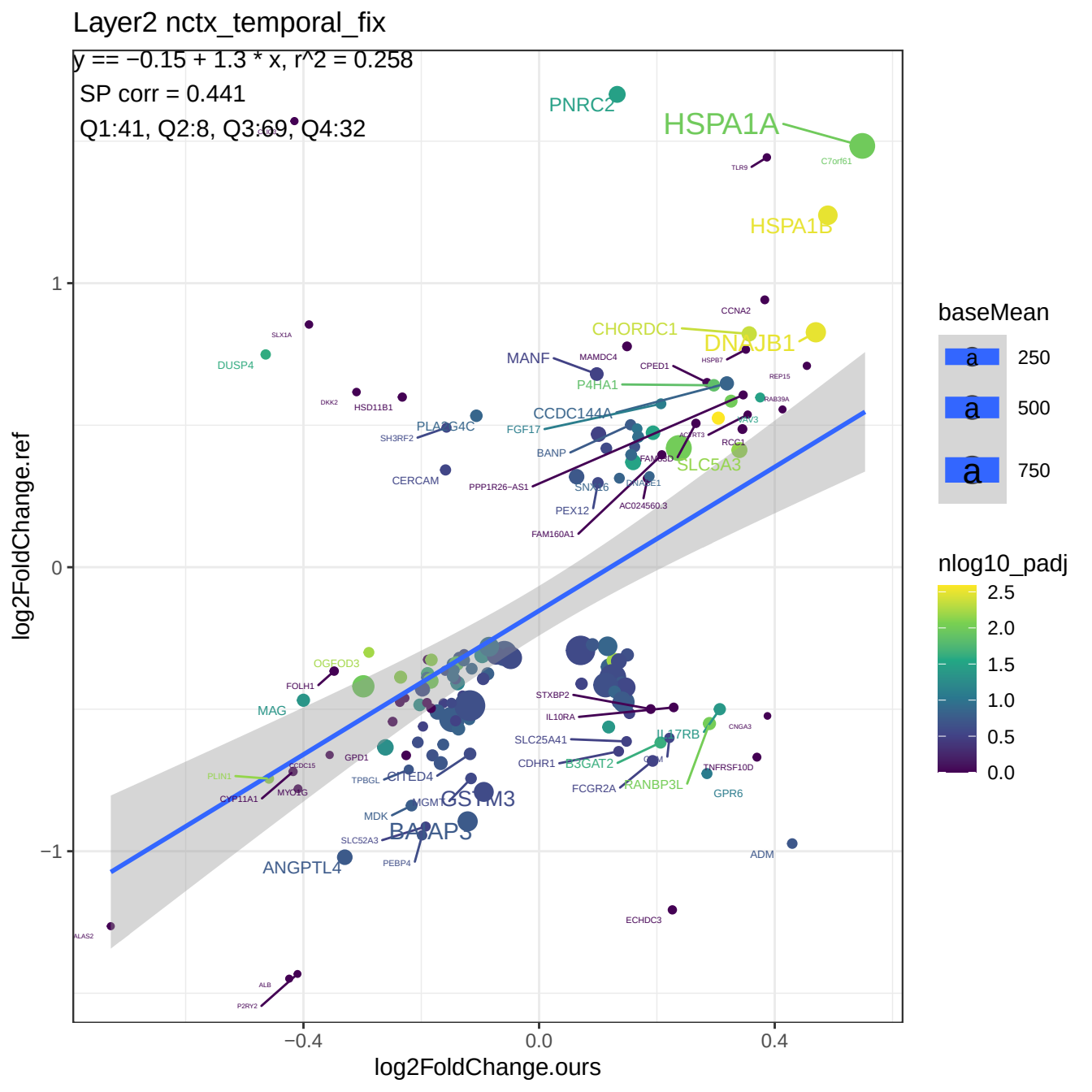
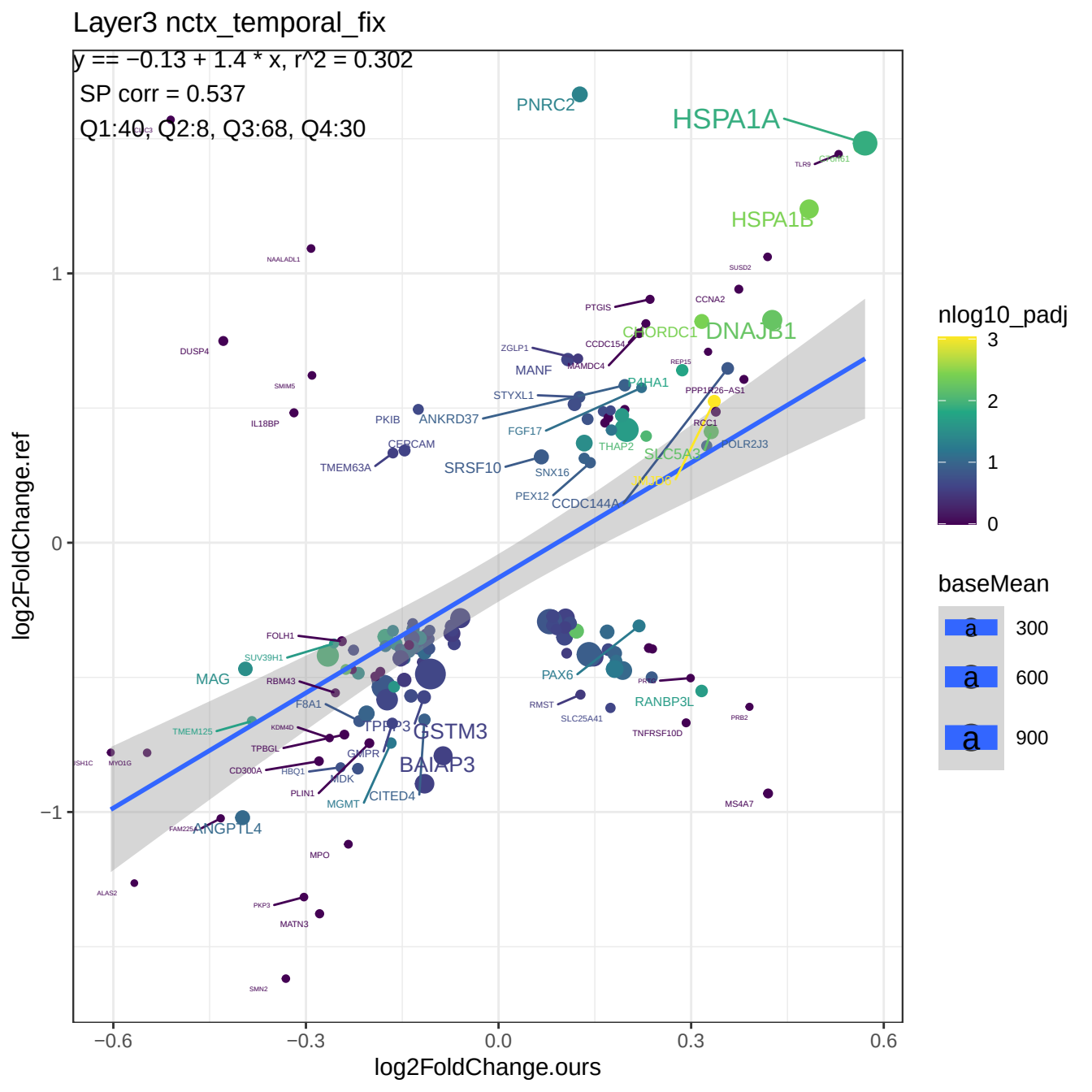
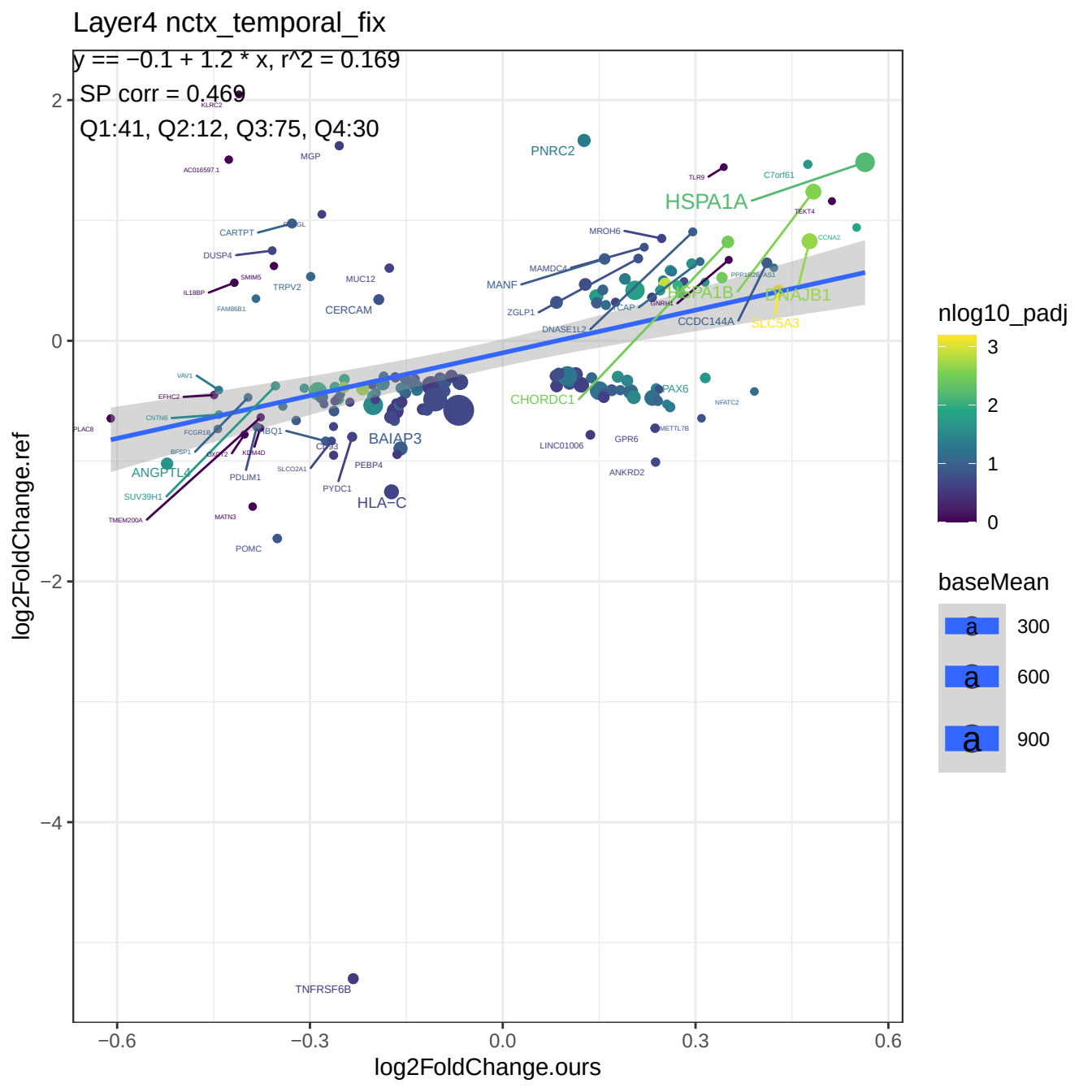
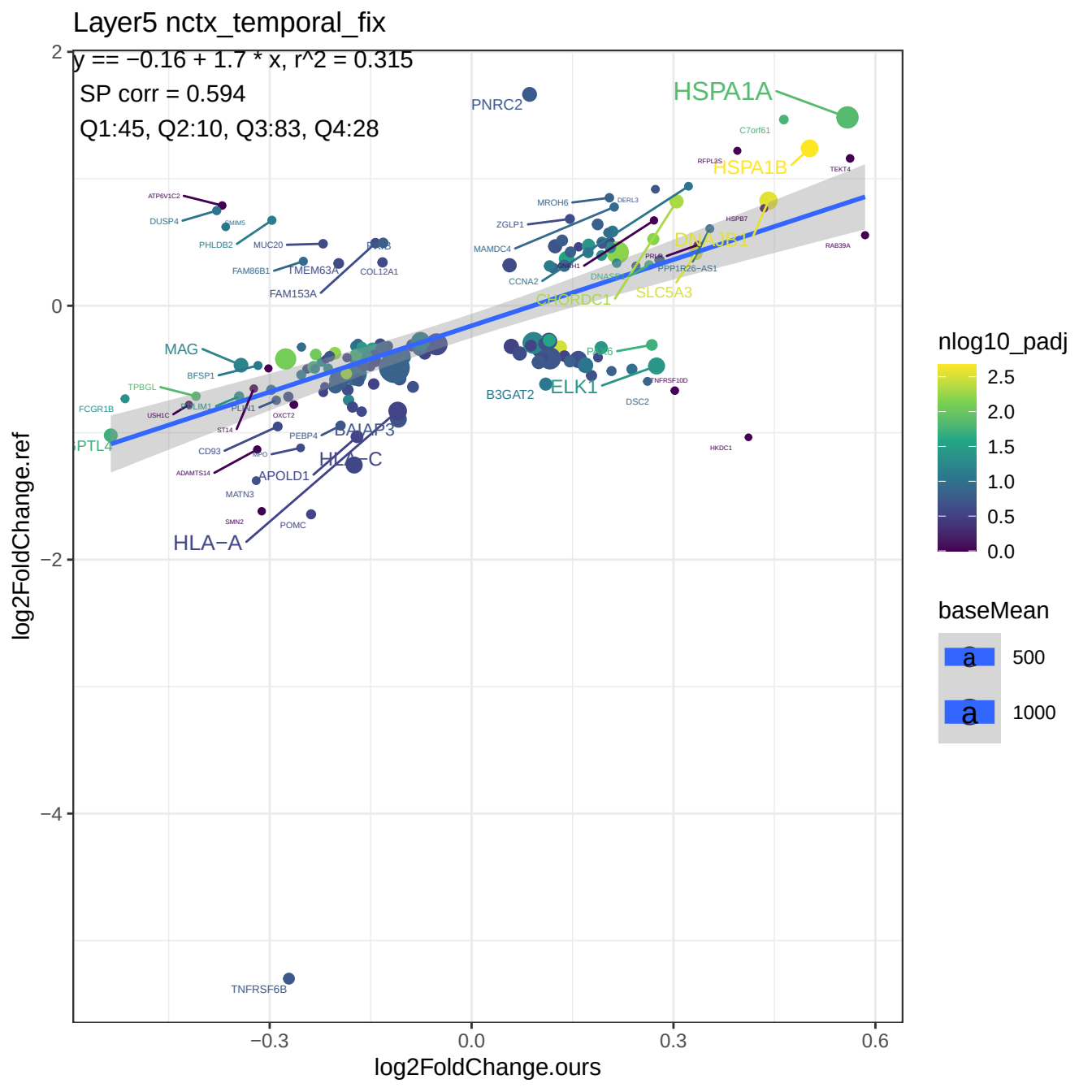
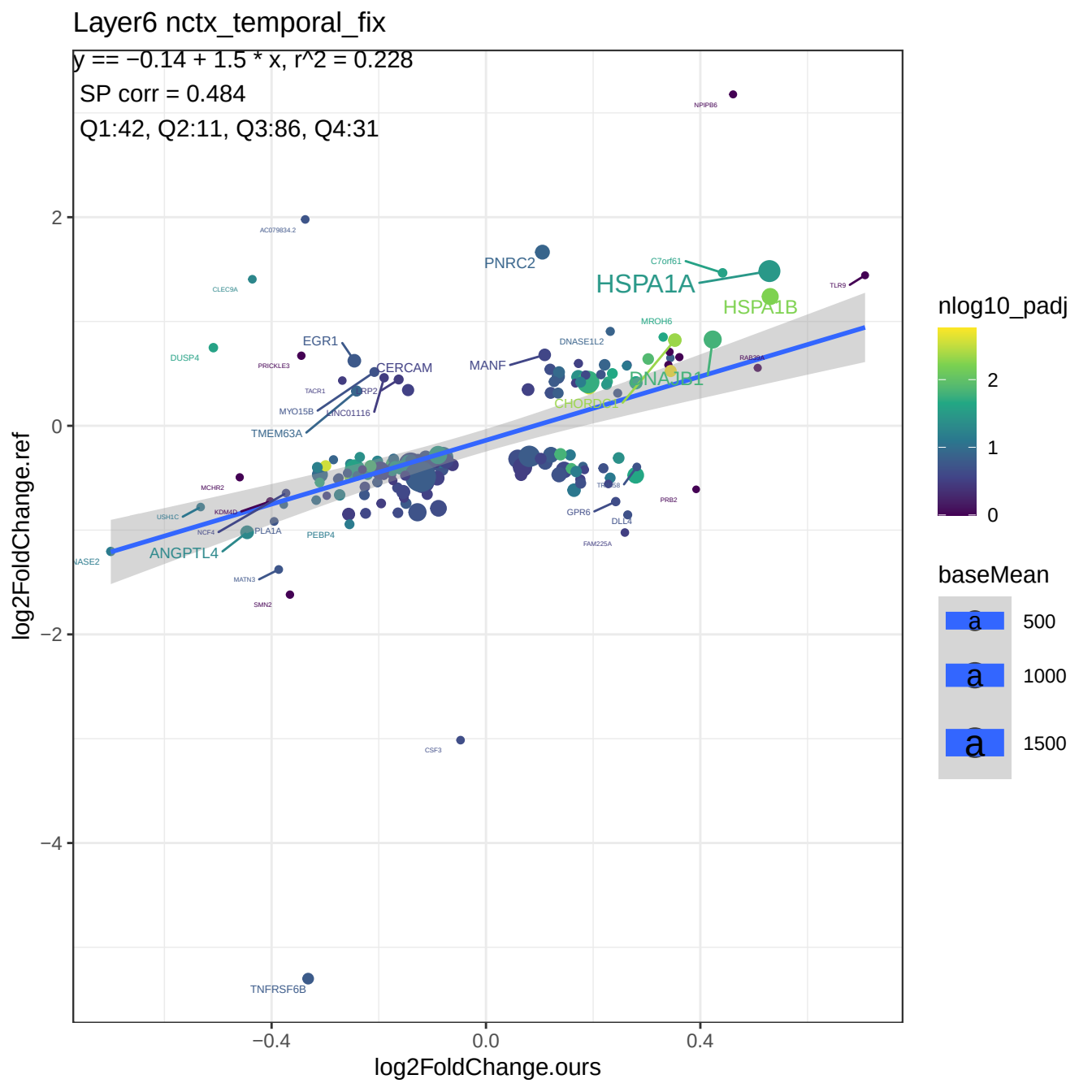

Layer1 nctx_temporal_fix
y == −0.27 + 1.5 * x, r^2 = 0.36
SP corr = 0.586
Q1:24, Q2:1, Q3:42, Q4:12
HSPA1A
C7orf61
HSPA1B
LYVE1
CHORDC1
CCNA2
RBM14
P4HA1
DNAJB9
LRP2
PLA2G4C
CDK2AP2
FGF17
HSPA4L
ANKRD37
DNASE1
VAV3
SPTSSB
NFIB
LIFR−AS1
NADK2
ROBO4
TLR1
IL17RB
ELK1
ADPRH
CCDC88C
IL4R
GEM
TPBGL
MGMT
RANBP3L
GPR6
ADM
SLCO2A1
HLA−A
PLIN1
SLC38A5
ANGPTL4
ANKRD2
HLA−C
DMRT3
TNFRSF6B
0
−2
−4
−0.5
0.0
0.5
log2FoldChange.ref
log2FoldChange.ours
nlog10_padj
1.00
0.75
0.50
0.25
0.00
baseMean
a
100
200
300
400

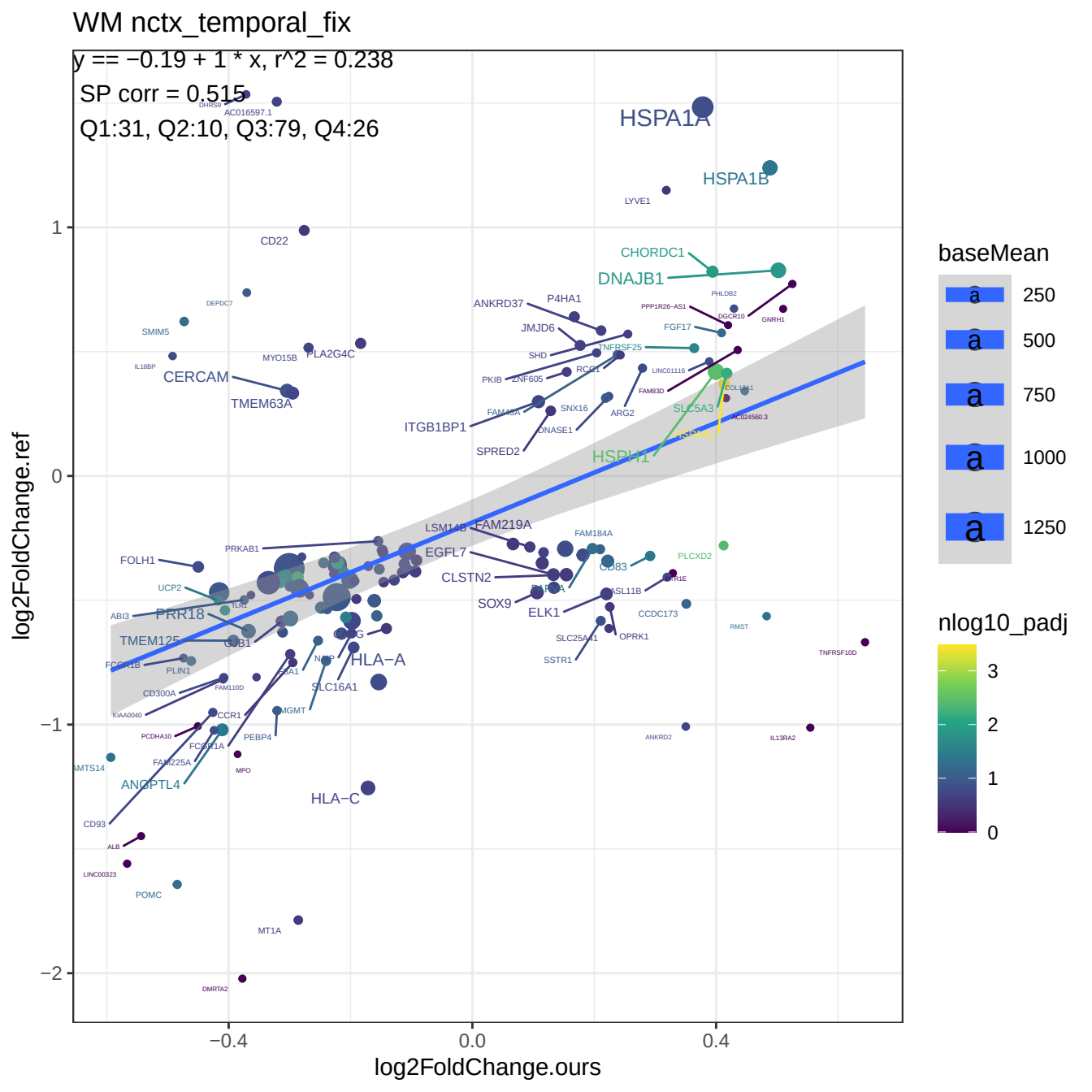

Layer2 nctx_temporal_fix
y == −0.15 + 1.3 * x, r^2 = 0.258
SP corr = 0.441
Q1:41, Q2:8, Q3:69, Q4:32
log2FoldChange.ref
log2FoldChange.ours
1
0
−1
−0.4
0.0
0.4
PNRC2
HSPA1A
C7orf61
TLR9
HSPA1B
CCNA2
CHORDC1
DNAJB1
SLX1A
DUSP4
MANF
MAMDC4
CPED1
HSPB7
REP15
P4HA1
RAB39A
DKK2
HSD11B1
CCDC144A
PLAAG4C
FGF17
SH3RF2
RCC1
BANP
SLC5A3
CERCAM
PPP1R26−AS1
SNAI2
PEX12
AC024560.3
FAM160A1
OGFOD3
FOLH1
MAG
STXBP2
IL10RA
CNGA3
SLC25A41
IL17RB
CDHR1
B3GAT2
TNFRSF10D
FCGR2A
RANBP3L
GPR6
PLIN1
CYP11A1
MYO1G
GPD1
TPBGL
CITED4
GSTM3
MDK
BAIAP3
SLC52A3
PEBP4
ANGPTL4
ADM
ECHDC3
ALAS2
ALB
P2RY2
baseMean
250
500
750
nlog10_padj
2.5
2.0
1.5
1.0
0.5
0.0


Layer3 nctx_temporal_fix
y == −0.13 + 1.4 * x, r^2 = 0.302
SP corr = 0.537
Q1:40, Q2:8, Q3:68, Q4:30
log2FoldChange.ref
log2FoldChange.ours
nlog10_padj
baseMean
PNRC2
HSPA1A
HSPA1B
DNAJB1
CCNA2
PTGIS
ZGLP1
CCDC154
MANF
MAMDC4
P4HA1
STYXL1
PPP1R26-AS1
PKIB
ANKRD37
FGF17
THAP2
SLC5A3
FOLR2J3
RCC1
CERCAM
TMEM63A
SRSF10
SNX16
PEX12
CCDC144A
NAALADL1
DUSP4
SMIM5
IL18BP
SUSD2
TLR9
FOLH1
SUV39H1
MAG
RBM43
F8A1
TMEM125
KDM4D
TPBGL
GMPR
CD300A
HBQ1
MDK
PLIN1
MGMT
TPPP3
GSTM3
BAIAP3
CITED4
ANGPTL4
MPO
ALAS2
PKP3
MATN3
SMN2
PAX6
RMST
SLC25A41
RANBP3L
TNFRSF10D
MS4A7
PRB2
1
0
−1
−0.6
−0.3
0.0
0.3
0.6
3
2
1
0
300
600
900


Layer4 nctx_temporal_fix
y == −0.1 + 1.2 * x, r^2 = 0.169
SP corr = 0.469
Q1:41, Q2:12, Q3:75, Q4:30
log2FoldChange.ref
log2FoldChange.ours
2
0
−2
−4
−0.6
−0.3
0.0
0.3
0.6
PNRC2
HSPA1A
HSPA1B
DNAJB1
CCDC144A
SLC5A3
C7orf61
TEKT4
CCNA2
TLR9
MROH6
MAMDC4
MANF
ZGLP1
DNASE1L2
CHORDC1
PAX6
NFATC2
METTL7B
GPR6
LINC01006
ANKRD2
BAIAP3
PEBP4
PYDC1
HLA−C
ANGPTL4
SUV39H1
TMEM200A
PDLIM1
SLCO2A1
MATN3
POMC
TNFRSF6B
CARTPT
DUSP4
SMIM5
IL18BP
TRPV2
MUC12
CERCAM
FAM86B1
MGP
AC016597.1
KLRC2
VAV1
EFHC2
CNTN6
FCGR1B
PLAC8
BFSP1
KDM4D
nlog10_padj
3
2
1
0
baseMean
a
300
a
600
a
900


Layer5 nctx_temporal_fix
y == −0.16 + 1.7 * x, r^2 = 0.315
SP corr = 0.594
Q1:45, Q2:10, Q3:83, Q4:28
log2FoldChange.ref
log2FoldChange.ours
2
0
−2
−4
−0.3
0.0
0.3
0.6
PNRC2
HSPA1A
C7orf61
RFPL2S
HSPA1B
TEKT4
ATP6V1C2
DUSP4
SMIM5
PHLDB2
MUC20
FAM86B1
TMEM63A
FAM153A
COL12A1
MROH6
DERL3
ZGLP1
HSPB7
MAMDC4
DNAJB1
RAB39A
CCNA2
PRR5
PPP1R26−AS1
DNASE2
CHORDC1
SLC5A3
MAG
BFSP1
TPBGL
FCGR1B
USH1C
PLIN1
OXCT2
ST14
CD93
ADAMTS14
PEBP4
BAIAP3
HLA−C
APOLD1
MATN3
SMN2
POMC
HLA−A
TNFRSF6B
B3GAT2
ELK1
DSC2
TNFRSF10D
HKDC1
nlog10_padj
2.5
2.0
1.5
1.0
0.5
0.0
baseMean
500
1000


Layer6 nctx_temporal_fix
y == −0.14 + 1.5 * x, r^2 = 0.228
SP corr = 0.484
Q1:42, Q2:11, Q3:86, Q4:31
log2FoldChange.ref
log2FoldChange.ours
2
0
−2
−4
−0.4
0.0
0.4
NPIPB6
AC079834.2
CLEC9A
PNRC2
C7orf61
HSPA1A
HSPA1B
TLR9
MROH6
DUSP4
EGR1
PRICKLE3
CERCAM
MANF
DNASE1L2
RAB39A
DNAJB1
TACR1
MYO15B
LINC01116
CHORDC1
TMEM63A
MCHR2
USH1C
KDM4D
NCF4
PLA1A
PEBP4
ANGPTL4
GPR6
PRB2
DLL4
FAM225A
MATN3
SMN2
CSF3
TNFRSF6B
nlog10_padj
2
1
0
baseMean
500
1000
1500


WM nctx_temporal_fix
y == −0.19 + 1 * x, r^2 = 0.238
SP corr = 0.515
Q1:31, Q2:10, Q3:79, Q4:26
log2FoldChange.ref
log2FoldChange.ours
1
0
−1
−2
−0.4
0.0
0.4
HSPA1A
HSPA1B
LYVE1
CD22
CHORDC1
DNAJB1
PHLDB2
PPP1R26−AS1
DGCR10
GNRH1
ANKRD37
P4HA1
JMJD6
FGF17
SHD
TNFRSF25
PKIB
ZNF605
RCC1
LINC01116
FAM83D
CERCAM
TMEM63A
MYO15B
PLA2G4C
SMIM5
DEPDC7
IL18BP
ITGB1BP1
SNX16
ARG2
SLC5A3
SPRED2
DNASE1
HSPH1
AC024560.3
FOLH1
PRKAB1
LSM14B
FAM219A
FAM184A
EGFL7
CD83
PLCXD2
CLSTN2
SOX9
ELK1
ASL11B
CCDC173
SLC25A41
OPRK1
SSTR1
RMST
TNFRSF10D
ANKRD2
IL13RA2
UCP2
ABI3
PRR18
TMEM125
FCGR1B
PLIN1
CD300A
KIAA0040
FAM110D
CCR1
PCDHA10
FCGR1A
FAM225A
MPO
PEBP4
SLC16A1
HLA−A
HLA−C
ANGPTL4
CD93
ALB
LINC00323
POMC
MT1A
DMRTA2
baseMean
250
500
750
1000
1250
nlog10_padj
3
2
1
0